Tokenomics & Future of Blockchain

Economics, Trends, and Opportunities

Suresh Paul

Algorithm Basics and BW

What We'll Cover Today

1. Bitcoin tokenomics deep dive
2. Ethereum tokenomics analysis
3. Supply and demand dynamics
4. Value drivers and price factors
5. Future of blockchain technology
6. Career opportunities in crypto/blockchain

Quick Recap: Real-World Applications

Last class we experienced:

- De-banking and crypto's solution
- Cross-border payments (crypto vs SWIFT)
- Tokenization of physical assets
- Security challenges and best practices
- Created MetaMask wallet + first transaction!

Today: Understanding crypto economics and future outlook

What is Tokenomics?

Definition:

- Token + Economics = Tokenomics
- Study of cryptocurrency supply and demand
- How tokens are created, distributed, and used
- Factors that affect token value

Why it matters:

- Understand what gives crypto value
- Compare different cryptocurrencies
- Make informed decisions
- Predict long-term viability

Key Tokenomics Concepts

1. **Total Supply:** Maximum number of tokens ever
2. **Circulating Supply:** Tokens available now
3. **Emission Rate:** How fast new tokens created
4. **Burn Rate:** How tokens are destroyed
5. **Distribution:** Who owns the tokens

Basic principle:

- More demand + Less supply = Higher price
- Less demand + More supply = Lower price

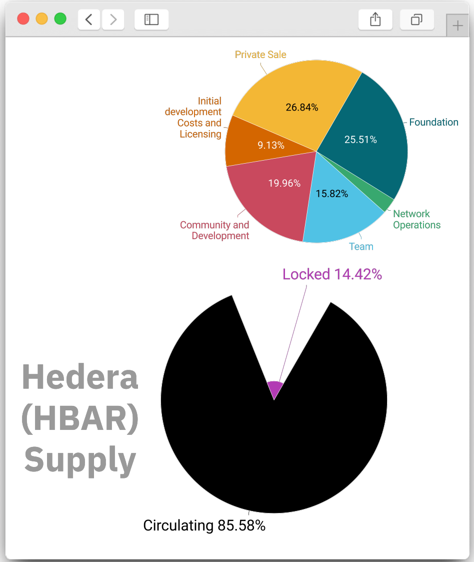
Supply Types Visualized

Three types of supply:

- ∞ **Max supply:** Hard cap
- **Circulating:** Currently available
- **Locked:** Held by founders, vesting schedule

Example: For Bitcoin ₿

- Max: 21M BTC
- Circulating: 19.6M BTC
- Unmined: 1.4M BTC



Bitcoin Tokenomics: Overview

Key facts:

- **Max Supply:** 21 million BTC (fixed forever)
- **Current Supply:** 19.6M BTC (93%)
- **Block Time:** 10 minutes
- **Block Reward:** 6.25 BTC (halves every 4 years)
- **Next Halving:** 2024 $\rightarrow$ 3.125 BTC reward

Key characteristic:

- Deflationary - Supply decreases over time
- Last Bitcoin mined around year 2140

Bitcoin Halving Explained

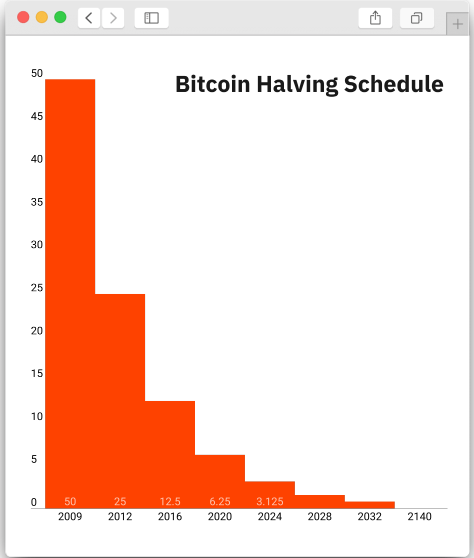
What is halving?

- Mining reward cut in half every 210,000 blocks (4 years)
- Reduces new supply entering market
- Programmed into Bitcoin code

Halving history:

1. 2009-2012: 50 ₿ per block
2. 2012-2016: 25 ₿ per block
3. 2016-2020: 12.5 ₿ per block
4. 2020-2024: 6.25 ₿ per block
5. 2024-2028: 3.125 ₿ per block

Bitcoin Supply Schedule



Why halving matters:

- 📉 Supply shock - Less new BTC
- 📈 Historically price increases
- 🕐 Predictable scarcity
- 💎 Mimics gold mining difficulty

Pattern:

- Each halving → Price tends to rise
- Not guaranteed, but historical trend
- Takes 6-18 months to see effect

Bitcoin Distribution

Who owns Bitcoin?

- Satoshi Nakamoto: 1M BTC (never moved)
- Companies: Tesla, MicroStrategy, Square, Algorithm Basics
- USA, China, India, El Salvador, Bhutan
- Exchanges: Binance, Coinbase holdings
- Retail investors: Millions of individuals
- Lost coins: 3-4M BTC (lost private keys)

Concentration:

- Top 1% addresses hold 90% of supply
- But many addresses = same entity (exchanges)
- Becoming more distributed over time

Bitcoin Value Drivers

What makes Bitcoin valuable?

1. **Scarcity:** Only 21M will ever exist
2. **Network effect:** More users = more valuable
3. **Security:** Most secure blockchain
4. **Adoption:** Growing acceptance worldwide
5. **Store of value:** "Digital gold" narrative
6. **Sentiment:** News and perception

Stock-to-Flow model:

- Ratio of existing supply to new supply
- Higher ratio = More scarce = Higher value (theory)

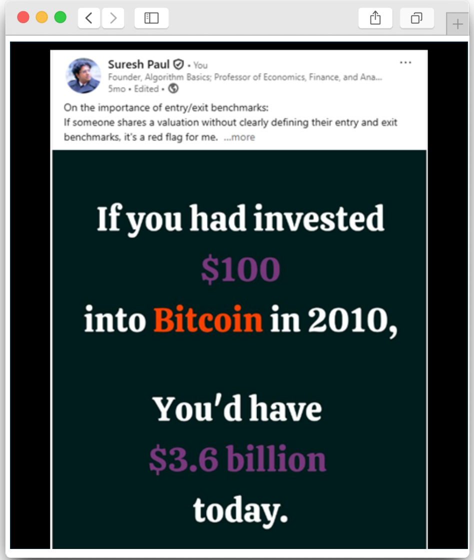
Bitcoin Price Cycles

Historical pattern:

- 📈 Bull market (3 years)
- 📉 Bear market (1 year)
- ♻️ Cycle repeats
- 📅 Often tied to halving

Past cycles:

- 2011: \$30 → \$2 → \$1,000
- 2013-2015: \$1,000 → \$200 → \$20k
- 2017-2018: \$20k → \$3k → \$69k
- 2024-2026: \$69k → \$16k → ?



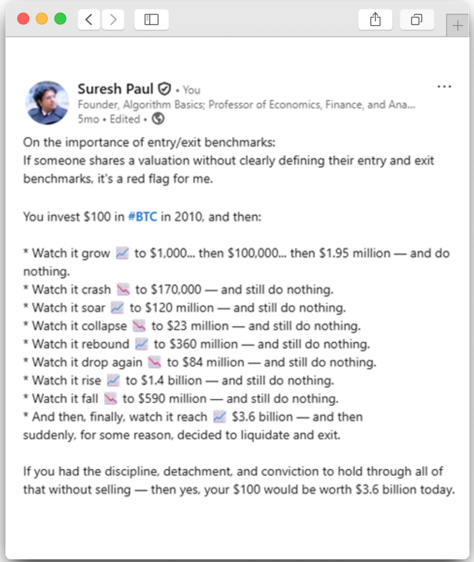
Bitcoin Price Cycles

Historical pattern:

- 📈 Bull market (3 years)
- 📉 Bear market (1 year)
- ♻ Cycle repeats
- 📅 Often tied to halving

Past cycles:

- 2011: \$30 $\rightarrow$ \$2 $\rightarrow$ \$1,000
- 2013-2015: \$1,000 $\rightarrow$ \$200 $\rightarrow$ \$20k
- 2017-2018: \$20k $\rightarrow$ \$3k $\rightarrow$ \$69k
- 2024-2026: \$69k $\rightarrow$ \$16k $\rightarrow$?



Suresh Paul · You
Founder, Algorithm Basics; Professor of Economics, Finance, and Ana...
5mo • Edited •

On the importance of entry/exit benchmarks:
If someone shares a valuation without clearly defining their entry and exit benchmarks, it's a red flag for me.

You invest $100 in #BTC in 2010, and then:

* Watch it grow 📈 to $1,000... then $100,000... then $1.95 million — and do nothing.
* Watch it crash 📉 to $170,000 — and still do nothing.
* Watch it soar 📈 to $120 million — and still do nothing.
* Watch it collapse 📉 to $23 million — and still do nothing.
* Watch it rebound 📈 to $360 million — and still do nothing.
* Watch it drop again 📉 to $84 million — and still do nothing.
* Watch it rise 📈 to $1.4 billion — and still do nothing.
* Watch it fall 📉 to $590 million — and still do nothing.
* And then, finally, watch it reach 📈 $3.6 billion — and then suddenly, for some reason, decided to liquidate and exit.

If you had the discipline, detachment, and conviction to hold through all of that without selling — then yes, your $100 would be worth $3.6 billion today.

Ethereum Tokenomics: Overview

Key facts:

- **Max Supply:** No hard cap (inflationary)
- **Current Supply:** 120M ETH
- **Block Time:** 12 seconds
- **Issuance:** 0.5% per year (after Merge)
- **Burn:** Transaction fees destroyed (EIP-1559)

Key characteristic:

- Potentially deflationary - Burns can exceed issuance
- Supply depends on network usage

Ethereum's Major Upgrade: The Merge

What happened in September 2022?

- Switched from Proof-of-Work to Proof-of-Stake
- Issuance reduced by 90%
- Energy consumption down 99.95%
- Staking rewards instead of mining

Impact on tokenomics:

1. Less new ETH created (supply shock)
2. Fee burning continues (EIP-1559)
3. Net result: Potentially deflationary

EIP-1559: Fee Burning Mechanism

How it works:

1. User sends transaction with base fee
2. 🔥 Base fee is burned (destroyed)
3. Optional tip goes to validators
4. More usage = More ETH burned

Effect on supply:

- High network activity $\rightarrow$ More burn
- Can burn more than issued (deflationary periods)
- Low activity $\rightarrow$ Slightly inflationary
- Net: 0% to -2% per year

Ethereum Supply Dynamics

Triple Halving:

- Pre-Merge: 4.3% inflation
- Post-Merge: 0.5% inflation
- 90% reduction
- With burn: Net deflationary

Comparison:

- Bitcoin: Predictable deflation
- Ethereum: Dynamic supply
- Gold: 2% inflation
- USD: 2-8% inflation

Ethereum Value Drivers

What makes Ethereum valuable?

1. **Utility:** Powers DeFi, NFTs, dApps
2. **Network effects:** Most developers
3. **Burn mechanism:** Deflationary pressure
4. **Staking yield:** 4% APR for holders
5. **Ecosystem:** Largest DeFi TVL
6. **Upgrades:** Continuous improvement

Ultrasound money:

- Community meme: ETH more deflationary than BTC
- Depends on network activity

Bitcoin vs Ethereum: Tokenomics Comparison

Feature	Bitcoin (BTC)	Ethereum (ETH)
Max Supply	21M (hard cap)	No cap
Current Supply	19.6M	120M
Issuance Rate	Halves every 4 years	0.5% per year
Supply Model	Deflationary	Dynamic (inflation/deflation)
Burn Mechanism	No	Yes (EIP-1559)
Primary Use	Store of value	Platform/utility
Consensus	Proof-of-Work	Proof-of-Stake

Both are considered scarce relative to fiat currencies

Supply and Demand: The Big Picture

Supply factors:

- New coins issued (inflation)
- Coins burned (deflation)
- Coins locked (staking, vesting)
- Lost coins (forgotten wallets)

Demand factors:

- User adoption
- Institutional investment
- Media attention
- Speculation
- Use cases and utility

Market Cycles and Psychology

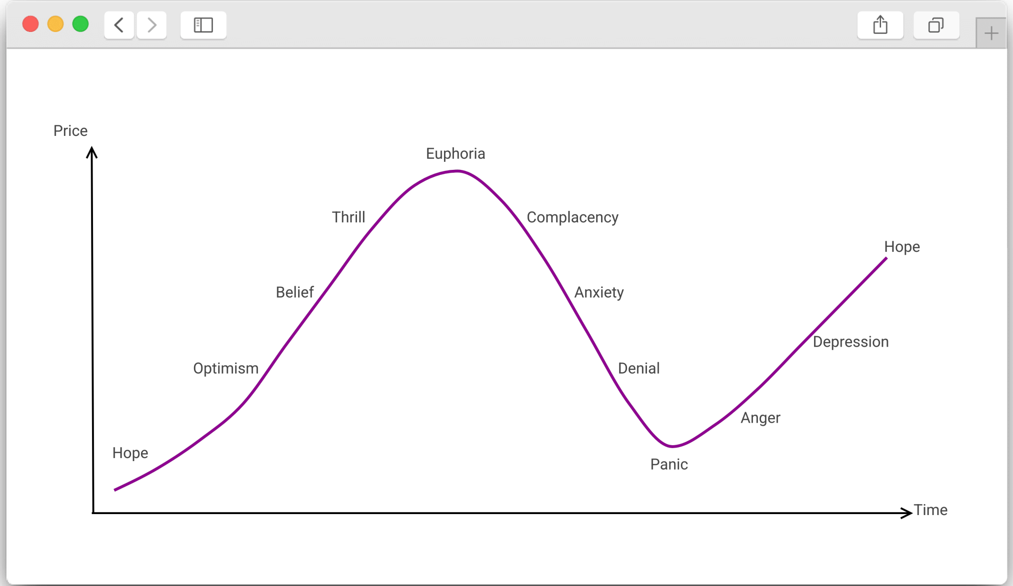
Crypto market psychology:

- **Euphoria:** Everyone buying
- **Panic:** Everyone selling
- **Cycles repeat**
- Driven by emotion

Stages:

1. Disbelief
2. Hope
3. Optimism
4. Belief
5. Thrill
6. Euphoria $\leftarrow$ Peak

Psychology of a Market Cycle



Other Tokenomic Models

Beyond BTC and ETH:

1. **Stablecoins:** Pegged to an existing asset (USDC, DAI, EURC, PAXG)
2. **Governance tokens:** Voting rights (UNI, AAVE, APE)
3. **Buyback & burn:** Company buys and destroys (BNB, HYPE, OP)
4. **Staking rewards:** Lock tokens, earn yield (ATOM, SOL, HBAR)
5. **Fair launch:** No pre-mine (BTC, TEZOS, FIL)

Red flags in tokenomics:

- Huge team allocation (>20%)
- Unlimited supply with no burn
- Complex vesting that enriches founders

Future of Blockchain: Technology Trends

What's coming next?

1. **Scalability:** Layer 2 solutions (Lightning, Polygon)
2. **Speed:** Faster blockchains (ICP, Solana, Avalanche)
3. **Privacy:** Zero-knowledge proofs (ZK-rollups, POL/MATIC)
4. **Interoperability:** Cross-chain bridges (LINK, QNT)
5. **Security:** Quantum-resistant cryptography (ALGO, *HBAR*)

Future of Blockchain: Technology Trends

Key innovation areas:

- Making crypto easier to use – DeFi, DeSci, DePIN
- Reducing transaction costs – Cross border payments, Tokenization
- Improving user experience – Gaming

Layer 2 Solutions

What are Layer 2s?

- Built on top of Layer 1 (Ethereum, Bitcoin)
- Process transactions faster and cheaper
- Inherit security from base layer
- Send final state back to Layer 1

Examples:

- Lightning Network (Bitcoin) - Instant payments
- Polygon (Ethereum) - Faster, cheaper transactions
- Arbitrum, Optimism (Ethereum) - Rollups

Result: 100x cheaper fees, 1000x more throughput

Blockchain Layers Explained

Layer 1:

- Base blockchain
- Most secure
- Slower, expensive

Layer 2:

- Fast transactions
- $ Low fees
- Settles to L1

Applications:

- User interface
- Built on L2 or L1

Adoption Trends

Who's adopting blockchain?

1. 🏢 **Enterprises:** IBM, Microsoft, JPMorgan
2. 🏦 **Banks:** SWIFT exploring blockchain
3. 🏛 **Governments:** Digital currencies (CBDCs)
4. 🏪 **Retail:** Walmart supply chain tracking
5. 🏛 **Education:** MIT, Stanford research

Growth metrics:

- 420M+ crypto users globally, $2T+ market cap (2024)
- Growing 20-30% year over year

Central Bank Digital Currencies (CBDCs)

What are CBDCs?

- 🏛️ Digital version of national currency
- 🏛 Issued by central banks
- ⛓️ May use blockchain technology
- 📱 Digital wallet access for all

Status:

1. China: Digital Yuan live
2. USA: Digital Dollar research
3. 90+ countries exploring

Different from crypto: Centralized, controlled by government

Regulatory Landscape

Regulations evolving globally:

- **USA:** SEC classifying tokens, upcoming clarity
- € **EU:** MiCA regulation (2024) - comprehensive framework
- ¥ **Asia:** Japan friendly, China restrictive
- **Global:** Moving toward regulation, not bans

Trend:

- More clarity = More institutional adoption
- Balance innovation with consumer protection
- Crypto maturing into regulated financial system

Future Predictions and Possibilities

Conservative predictions (5-10 years):

1. 🏛 Most banks will offer crypto services
2. Major retailers accept cryptocurrency
3. Digital identity on blockchain standard
4. Supply chains transparently tracked
5. Gaming economies primarily on-chain

Ambitious predictions:

- Traditional finance migrates largely on-chain
- National currencies backed by crypto
- Most contracts are smart contracts

Potential Challenges Ahead

Technical challenges:

- Scalability limits
- User experience complexity
- Interoperability issues
- Energy consumption (PoW chains)

Social/Political:

- Regulatory uncertainty
- Scams and fraud
- Education needed
- Decentralization vs efficiency

Overcoming these = Mainstream adoption

Career Opportunities in Blockchain

High-demand roles:

1. **Blockchain Developer:** Smart contracts, dApps (\$100k-200k+)
2. **Security Auditor:** Find vulnerabilities (\$120k-250k+)
3. **Crypto Analyst:** Research and analysis (\$80k-150k)
4. **Business Development:** Partnerships (\$90k-180k)
5. **Compliance/Legal:** Regulatory expertise (\$100k-200k)

Skills needed:

- Programming
- Understanding of cryptography
- Business acumen

Career Opportunities in Blockchain (cont'd.)

1. **Content Creation:** Education, writing
2. **Community Management:** Build coder base and user base
3. **Product Management:** Strategy and roadmap
4. **Sales:** Enterprise blockchain solutions

MIS graduates well-positioned:

- Understand tech + business
- Bridge between technical and business teams
- Process improvement expertise valuable

Investment Considerations

- $ Only invest what you can afford to lose
- Diversify across different assets
- Long-term perspective (3-10 years)
- Do your own research (DYOR)
- Understand volatility is normal
- Treat it as learning, not gambling

Key Takeaways

1. Bitcoin: Fixed supply, deflationary, digital gold
2. Ethereum: Dynamic supply, utility-driven, platform
3. Tokenomics = Supply + Demand + Utility
4. Future: Scaling, adoption, integration with traditional finance
5. Careers: Many opportunities, technical and non-technical
6. Keep learning: Technology evolving rapidly

Final thought:

- Blockchain is still early
- Understanding it now = competitive advantage

Questions and Discussion